Comfort in Sight

Optimizing Green Space Layouts using Genetic Algorithm

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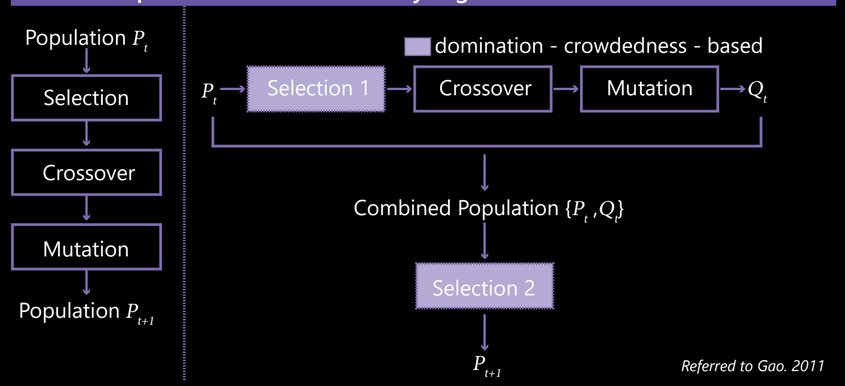
OVERVIEW OF NON-DOMINATED GENETIC ALGORITH II

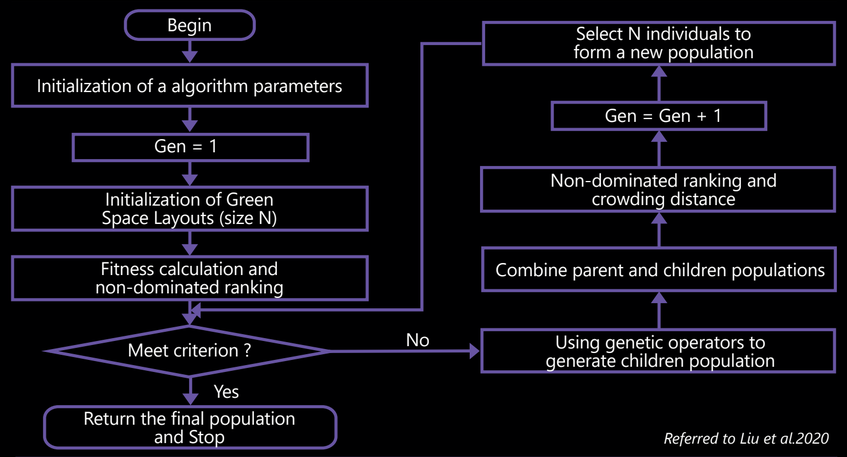
Constructing NSGA- II Optimisation Algorithm Logic

Comparison Between Ordinary Algorithm and NSGA-II



Referred to Gao. 2011

Flowchart of NSGA-II



Referred to Liu et al.2020

NSGA-II Second Selection Strategy

// Step 1: Non-dominated sorting

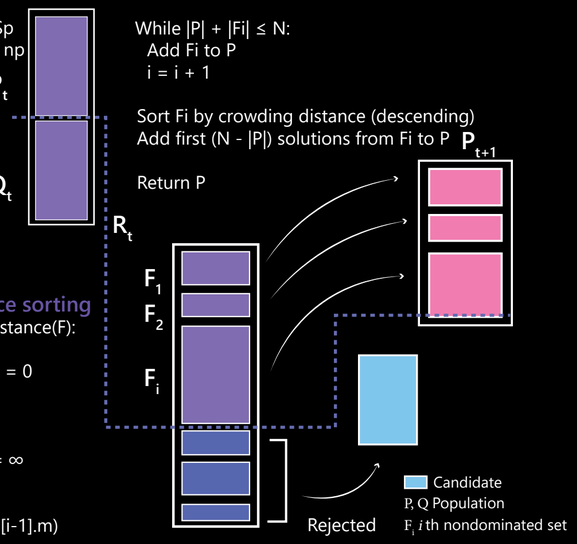
```
Function Fast-Non-Dominated-Sort(R):
  Initialize empty fronts F1, F2, ...
  For each solution p in R:
    For each solution q in R:
      If p dominates q: Add q to Sp
      If q dominates p: Increment np
    If np = 0: Add p to F1

  i = 1
  While Fi not empty:
    For each p in Fi:
      For each q in Sp:
        Decrement nq
        If nq = 0: Add q to Fi+1
    i = i + 1
  Return all fronts F1, F2, ...
```

// Step 2: Crowding distance sorting

```
Function Calculate-Crowding-Distance(F):
  n = |F|
  For each solution in F: distance = 0

  For each objective m:
    Sort F by objective m
    F[1].distance = F[n].distance = ∞

    For i = 2 to n-1:
      F[i].distance += (F[i+1].m - F[i-1].m)
```

// Step 3: Elite selection for next generation

```
Function Select-Next-Generation(F1, F2, ..., N):
  P = ∅
  i = 1

  While |P| + |Fi| ≤ N:
    Add Fi to P
    i = i + 1

  Sort Fi by crowding distance (descending)
  Add first (N - |P|) solutions from Fi to P

  Return P
```



NSGA-II Algorithm For Green Space Optimisation

Objectives:
1) Maximize Path Green View Index
2) Minimize Deviation from Target Green Area

// Main Workflow

```
// 1. Data Preparation
  greenSurfaces = CreateEffectiveGreenAreas(greenBoundaries, buildingFootprints)  // Boolean difference: green - buildings
  samplePoints = SampleAlongPaths(paths, 5.0)  // One sample every 5 meters
  grid = CreateUniformGrid(greenSurfaces)      // Divide green areas into grid cells

  // 2. Define Fitness Functions (CORE)
  function CalculateFitness(genes):  // Genes: Boolean array representing green state (true=green)
      // Objective 1: Green View Score (maximize)
      greenViewScore = CalculateGreenView(genes, grid, samplePoints, buildingFootprints)

      // Objective 2: Area Difference from Target (minimize)
      areaDifference = CalculateAreaDifference(genes, grid, targetArea)

      // Objective 3: Distribution Uniformity (minimize)
      distributionPenalty = CalculateDistributionPenalty(genes, grid)

      return [greenViewScore, areaDifference + distributionPenalty]
  }
  // 3. Optimize using NSGA-II
  paretoSet = RunNSGA2(CalculateFitness, grid.Count)
  bestSolution = SelectCompromiseSolution(paretoSet)

  return ConvertToGeometry(bestSolution, grid)
```

// Core Function 1: Green View Calculation

```
function CalculateGreenView(genes, grid, samplePoints, buildings):
  // Convert active genes to green geometry
  greenGeometry = []
  for i = 0 to genes.Length-1:
    if genes[i] == true:
      greenGeometry.Add(CreateRectangle(grid[i].centerPoint, gridSize))

  visibleGreenRayCount = 0
  totalRayCount = samplePoints.Count * 30  // 30 rays per sample point

  // Cast rays from each sample point to check green visibility
  foreach point in samplePoints:
    rays = GenerateRays(point, 30, 90.0)  // 30 rays with radius 90m

    foreach ray in rays:
      // Check if ray intersects green and is not blocked by buildings
      if RayIntersectsGreen(ray, greenGeometry) and
        !RayIsBlockedByBuildings(ray, buildings, greenGeometry):
        visibleGreenRayCount++

  return visibleGreenRayCount / totalRayCount  // Normalized score (0-1)
```

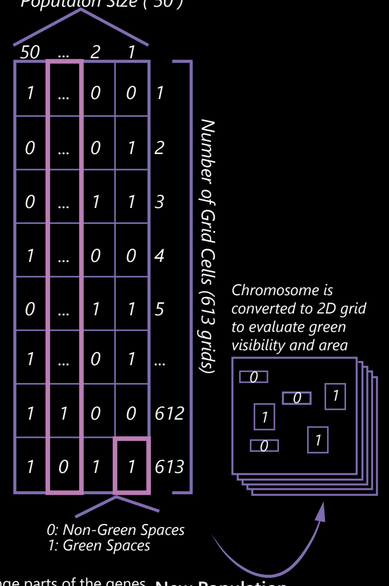
// Core Function 2: Area Difference Calculation

```
function CalculateAreaDifference(genes, grid, targetArea):
  activeCount = CountTrueValues(genes)
  actualArea = activeCount * gridSize * gridSize

  return |actualArea - targetArea|  // Absolute difference, smaller is better
```

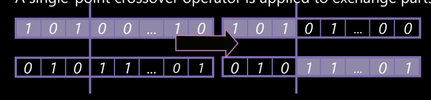
// Core Function 3: Distribution Penalty Calculation

```
function CalculateDistributionPenalty(genes, grid):
  // Calculate green ratio for each region
  regionGreenRatios = CalculateRegionGreenRatios(genes, grid)

  penalty = 0
  foreach region in regionGreenRatios:
    // Heavy penalty for regions with no green
    if region.ratio == 0:
      penalty += 1.0
    // Light penalty for regions with low green ratio
    else if region.ratio < 0.1:
      penalty += (0.1 - region.ratio) * 5.0

  return penalty
```

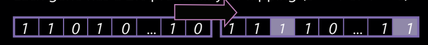
Representation



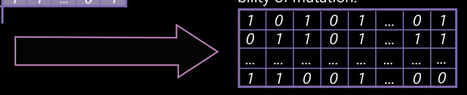
Crossover
Two binary chromosomes are selected randomly
A single-point crossover operator is applied to exchange parts of the genes



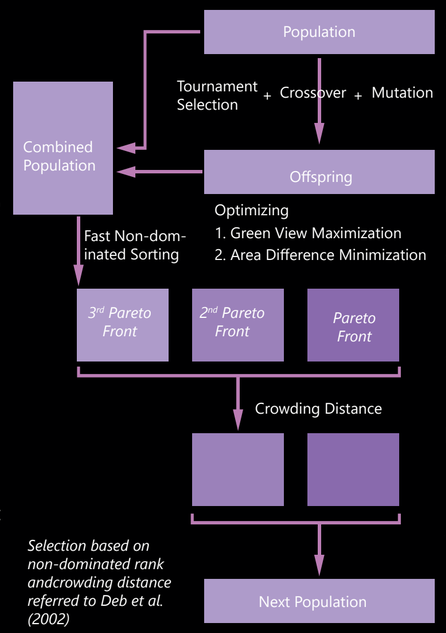
Mutation
One binary chromosome is selected randomly
Each gene has a 2% probability of flipping (0 → 1 or 1 → 0)



New Population
New individuals are 100% generated through crossover, with each gene having a 2% probability of mutation.



Selection Process



Selection based on non-dominated rank andcrowding distance referred to Deb et al. (2002)

// NSGA-II Algorithm Simplified

```
function RunNSGA2(fitnessFunction, geneLength):
  // Representation: Binary array (true/false) for grid cells
  population = GenerateRandomPopulation(100, geneLength)  // 100 solutions

  for generation = 1 to 50:  // Evolve for 50 generations
      // Selection: Binary tournament based on non-dominated ranking and crowding
      // Crossover: Uniform crossover exchanging genes between parents
      // Mutation: Low-probability bit-flip mutation changing individual grid states
      offspring = CreateOffspring(population)

      // Combine and sort by non-domination relationship
      combinedPopulation = population + offspring
      sortedPopulation = NonDominatedSort(combinedPopulation)

      // Select new generation
      population = SelectNewGeneration(sortedPopulation, 100)

  // Return Pareto-optimal solutions
  return GetParetoFront(population)
```


COMFORT IN SIGHT Created by: **Rui Gu** **BARC0034** Morphogenetic Programming

METHODOLOGY FRAMEWORK

Adapting Layouts through Visual Analysis and Evolutionary Logic

Two Optimisation Goals

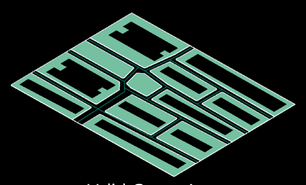
Green Space has been proved to be benefcial to human health especially to the elderly. Meanwhile, there is an increasing number of studies showing that the green visibility impacts the health of elderly greatly.Therefore, accoring to this, the project chose a residential area located in Guangdong Province, China as the research target.

This project is trying to promote data-driven and user-oriented urban design, particularly for aging societies, where green space configurations are no longer arbitrarily designed but intelligently optimized for well-being, inclusiveness, and long-term livability.

Stage 1: Site Analysis and Data Processing

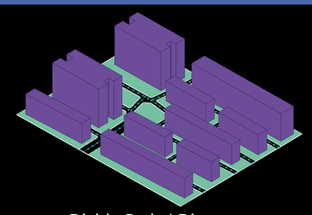
Green Boundaries Processing

Building Footprint Exclusion



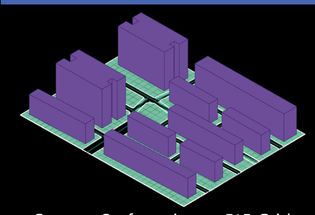
Valid Green Area

Paths' Sampling Points Generation (5m intervals)



Divide Paths' Distance

Valid Green Space Area Conversion to 5*5 Grid



Convert Surfaces into a 5*5 Grid

Stage 2: Grid Representation and Modelling

Greate Grid Units (gridRects)

Calculater Centers Points (gridCenters)

Use Boolean Array to Represent Green Space Status (gridStates: True: Green Space False:Non-Green Space)

Distributed across All Zones

Exclude the GreenAreas under Buildings

Generate Initial Population (Random Boolean Arrays)

Stage 3: Multi-Objective Optimization (NSGA-II)

Objective Function 1: Green Visibility Calculation

$$F_{GV} = 1/(n \cdot r) \sum_{i=1} n \sum_{k=1} r\, V(p_i, \theta_k)$$

$$V(p_i, \theta_k) = [\![G(p_i, \theta_k) \wedge \neg B(p_i, \theta_k)]\!]$$

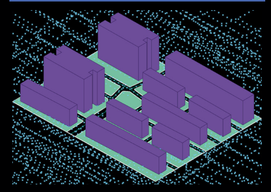
- F_{GV} = normalized green view index, ranging from [0, 1]
- n: number of sampling points
- r = number of rays emitted from each sampling point
- p_i: position of the i-th sampling point
- θ_k: angle of the k-th ray, $\theta k = 2\pi k/r$
- $G(P_i, \theta_k)$ = whether the ray intersects with green space
- $B(P_i, \theta_k)$ = whether a building blocks the green space in fornt of the ray
- $[\![\cdot]\!]$ = iversion bracket,which equals 1 if the condition inside is true, otherwise 0

Cast Rays from Sampling Points

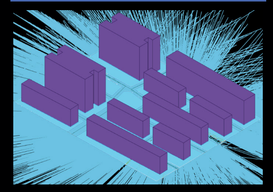
Ray Length 90-120 units

Detect Ray-Green Intersection

Building Obstruction Detection



Generate Isovist Points



Green Visibility Lines

Objective Function 2: Green Area Calculation

Actual Green Space Area Calculation

$$A_G = c^2 \cdot \Sigma_{i=1}^{m} g_i$$

Area Suitability Calculation

$$F_A = |A_G - A_T| + P_D$$

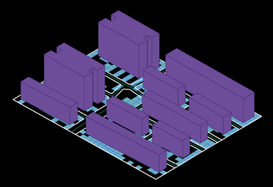
- A_G = total actual green space area
- c: size of each grid cells (cellSize)
- m = total number of grid cells
- g_i: status of the i-th grid cell (1 indicates green space, 0 indicates non-green space)
- F_A: area suitability value (the smaller, the better)
- A_T = target green space area (target area)
- P_D = penalty value for spatial distribution
- $|\cdot|$ = absolute value operator

Calculate Total Green Area

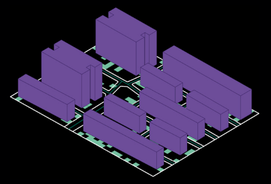
Compare with Target Area

Calculate Area Difference

Green Space Distribution Uniformity



Potential Green Area



Optimized Green Area

Pareto Optimal Solution Set: Maximize Green Visibility & Minimize Green Area Difference

ADAPTATION PROCESS

Algorithmic Thinking --- From Theory to Practice

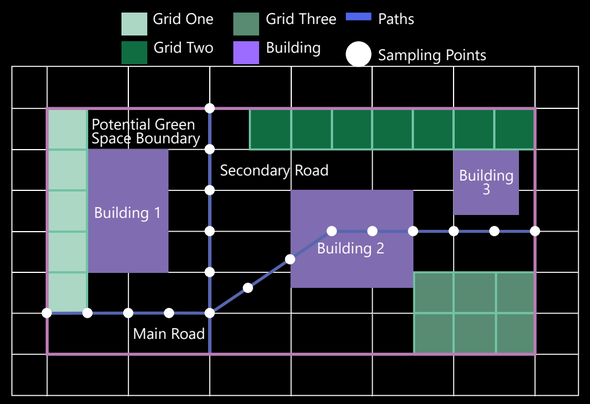
Environmental Adaptation Mechanism

Spatial Recognition and Constraints

Building Adaptation: Automatically identify areas relative to building footprints.
Path Sampling: Optimize sampling point generation along walking paths.
Regional Adaptation: Automatically recognize and differentiate multiple zones.

Green Visibility Evaluation

Raycasting Simulation: Simulate human vision through ray emission to detect green visibility.
Occlusion Consideration: Account for building occlusions impacting green view



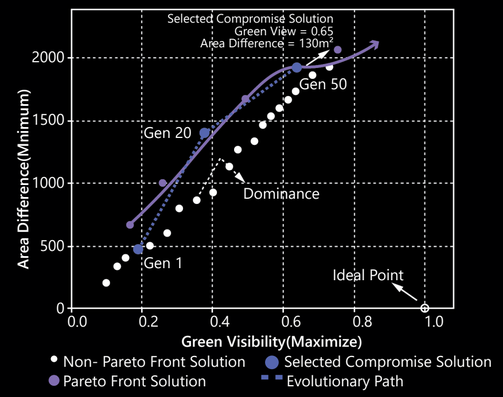
Multi-objective Optimization Process

NSGA-II Algorithm Implementation

Population Initialization: Generate initial green layouts randomly.
Multi-objective Fitness: Calculate green view gain and green area variance.
Regional Balancing: Use penalty factors to ensure green space distribution balance.
Non-dominated Sorting: Rank solutions based on Pareto optimality.

Trade-off Solution Selection

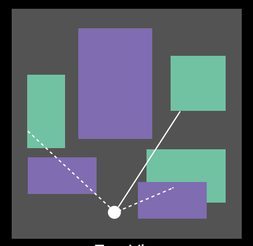
Calculate distance to ideal objectives and select a well-balanced solution.



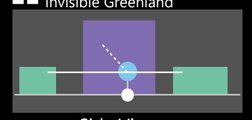
Green Visibility Evaluation Mechanism

Green view evaluation is the core adaptation mechanism, ensuring optimization results match human perceptual experiences.
Each sampling point emits rays to assess visibility and building occlusion impacts, providing a perceptual-based green view ratio.

F_{GV} = Normalized green view index, ranging from [0, 1]



Top View



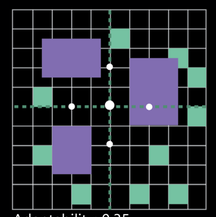
Side View

Greenland
Building
Sampling Points
Visible Greenland
Invisible Greenland

Green Space Layout Adaptation

Through iterative optimization, the green space layout gradually adapts to the spatial needs of the community environment.
From the initial random distribution to the final optimized state, the layout evolves into a more balanced pattern along pedestrian paths.
The system dynamically balances two competing objectives—green view improvement and area equality—meeting both perceptual experience needs and planning requirements.
Each region receives a reasonable allocation of green spaces, avoiding uneven distribution issues.

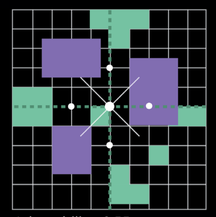
Initial Random Layout



Adaptability: 0.35
Green View Index: 0.35
Area Variance: 220 m²

Random initial distribution leads to scattered green spaces and low visibility

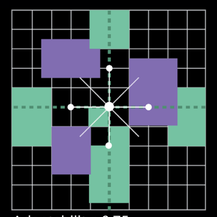
Intermediate Evolved Layout



Adaptability: 0.55
Green View Index: 0.55
Area Variance: 120 m²

After multiple generations, green spaces begin clustering around pathways, improving visibility

Final Optimized Layout



Adaptability: 0.75
Green View Index: 0.75
Area Variance: 60 m²

Final optimized layout achieves well-distributed green spaces along community paths, maximizing green view

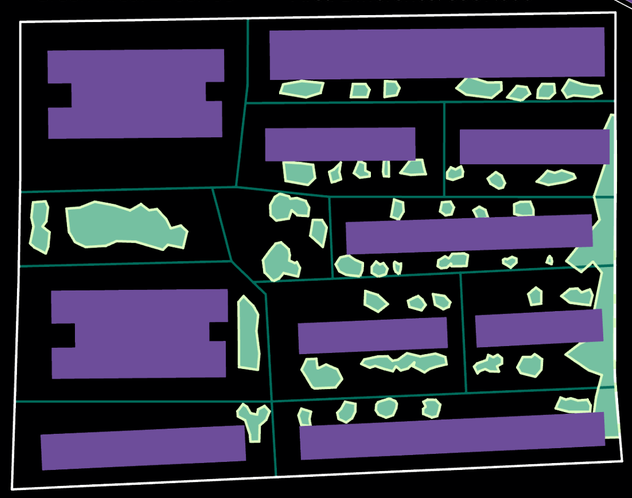
BEFORE AND AFTER OPTIMISATION

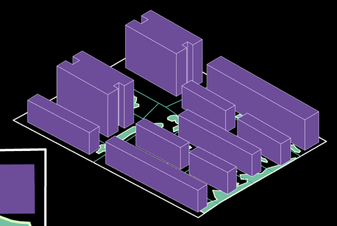
Check the Optimisation Results

Current status of the site

Green Visibility: 0.753
Green Area: 2488.193

Planned Area: 6476
Area Difference: 3987.806





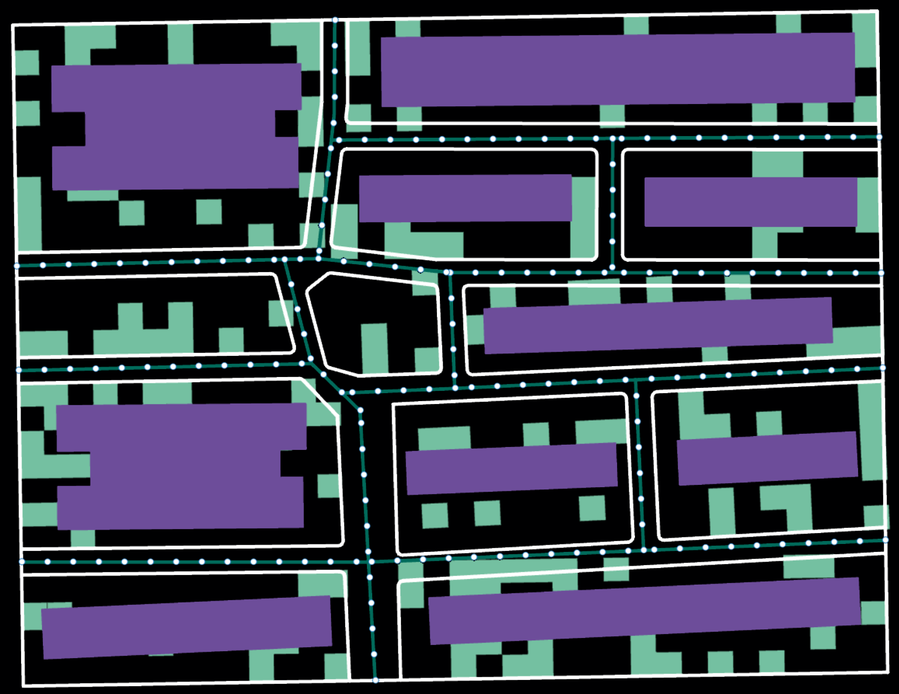
In the previous computer vision analysis of the perceived comfort of the elderly, it was found that the relationship between green visibility and the comfort of the elderly showed a positive. After analyzing the site, it was found that the residential area did not achieve a good level of comfort. The current situation of the site has scattered green space distribution in the form of patches, and is not evenly distributed on the path.

Optimized Green Space Layout

The optimized green space layout maximizes the visual exposure to greenery for pedestrians, improving environmental comfort and enhancing the walking experience within the community. The total green space area after optimization closely matches the planned target (6476 m^2), ensuring regulatory compliance without unnecessary land use or resource wastage.
What's more, the layout guarantees that each designated green area boundary contains a reasonable proportion of green space, avoiding underutilized or barren zones and promoting spatial equity.

Green Visibility: 0.972
Green Area: 5239.824263

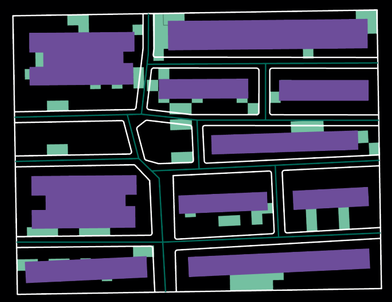
Planned Area: 6476
Area Difference: 1236.175



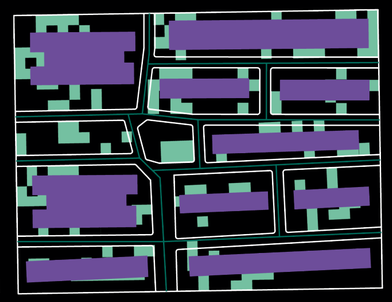
INBETWEEN GENERATED RESULTS COMPARISONS

Revealing the Optimal Solution of Comfort

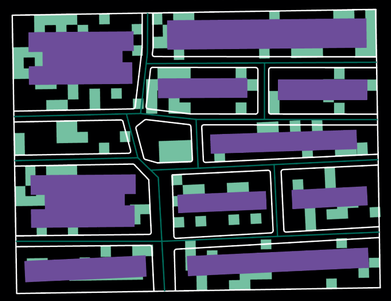
Green Visibility: 0.828
Green Area: 3295.243
Area Difference: 3180.756
N54



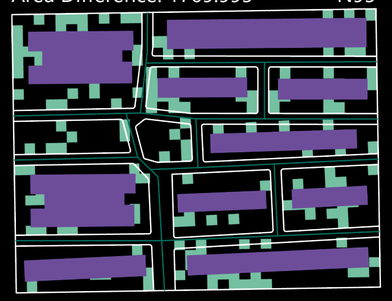
Green Visibility: 0.946
Green Area: 4276.302
Area Difference: 2199.697
N70



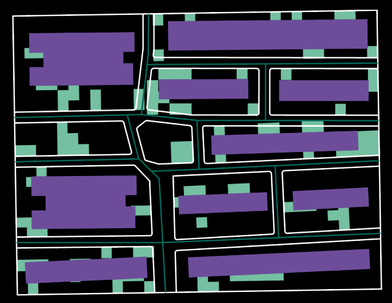
Green Visibility: 0.949
Green Area: 4718.413
Area Difference: 1757.5868
N90



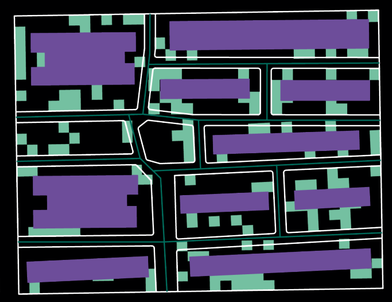
Green Visibility: 0.938
Green Area: 4706.004
Area Difference: 1769.995
N95



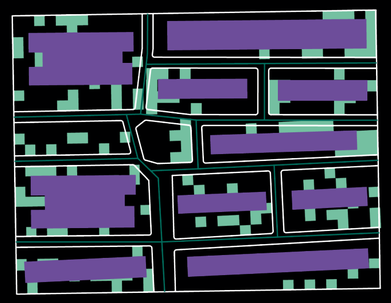
Green Visibility: 0.927
Green Area: 3776.050
Area Difference: 2699.949
N64



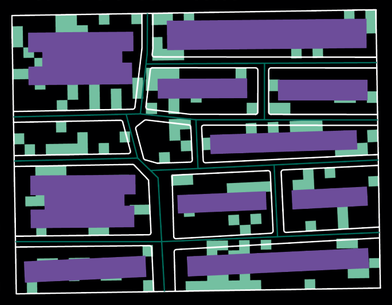
Green Visibility: 0.933
Green Area: 4685.672
Area Difference: 1790.327
N88



Green Visibility: 0.935
Green Area: 4875.501
Area Difference: 1600.498
N94

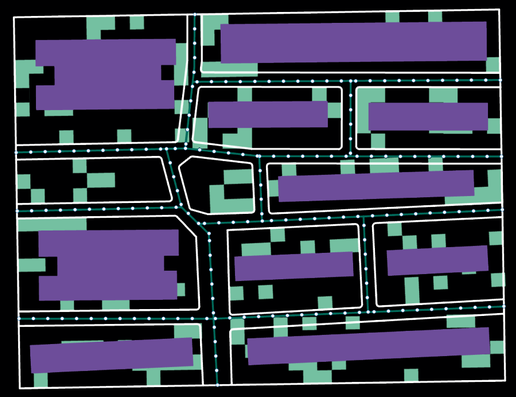


Green Visibility: 0.945
Green Area: 4686.348
Area Difference: 1789.651
N96

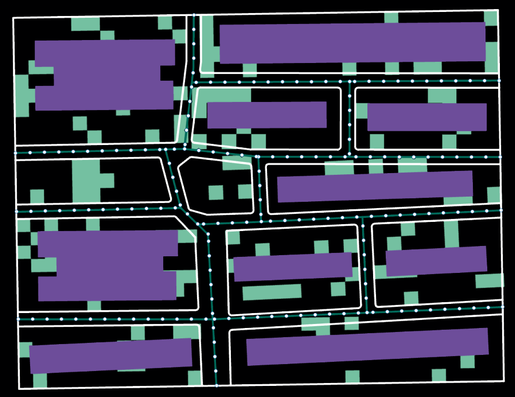


EVALUATE SOLUTIONS

Check the Optimisation Results



N97
Green Visibility: 0.943
Green Area: 4756.554
Area Difference: 1719.445



N104
Green Visibility: 0.963
Green Area: 4617.839
Area Difference: 1858.160

FUTURE WORK & LIMITATIONS

- **Lack of Parameter Calibration in View Analysis**
- **No Control Over Spatial Continuity or Shape of Green Areas**
- **No Integration with Environmental or Behavioral Data:**
  The optimization is static and does not respond to external data such as sunlight, wind, or pedestrian movement. Future work could integrate real-time or context-aware inputs to enable adaptive green space planning.
- **Limited Evaluation Dimensions:**
  The optimization currently considers only visual exposure and area compliance. Other important aspects—such as accessibility, usability, sunlight exposure, and acoustic buffering—are not yet incorporated into the fitness evaluation.